

Propositional eternalism versus propositional temporalism

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1. The “B-theory” versus “A-theory” debate: slogans

<i>The A theory</i>	<i>The B-theory</i>
The present time is “metaphysically special”, “privileged”, “distinguished”...	All times are “on a par”.
There are facts about which times are past, present and future, over and above the facts about which times are before which.	There are no such further facts. Pastness, presentness and futurity “reduce” to beforeness and afterness.
Time “flows” and “passes”. There is ‘objective becoming’.	Reality is a “block universe”.
The facts change.	Change is just a certain kind of pattern in the facts.

2. Propositional temporalism and propositional eternalism

Propositional eternalism: every true proposition is always true.

Or: $\forall p(p \rightarrow \text{Always } p)$

Propositional temporalism: some true propositions are not always true.

Or: $\exists p(p \wedge \neg \text{Always } p)$

Frege is the archetypal propositional eternalist:

But are there not thoughts which are true today but false in six months time? The thought, for example, that the tree there is covered with green leaves, will surely be false in six months time. No, for it is not the same thought at all. The words ‘This tree is covered with green leaves’ are not sufficient by themselves for the utterance, the time of utterance is involved as well. Without the time-indication this gives we have no complete thought, i.e. no thought at all. Only a sentence supplemented by a time-indication and complete in every respect expresses a thought. But this, if it is true, is true not only today or tomorrow but timelessly. (Frege, ‘The Thought’.)

3. Analogues

Propositional necessitism: every truth is necessarily true, $\forall p(p \rightarrow \Box p)$

Propositional contingentism: some truth is not necessarily true, $\exists p(p \wedge \neg \Box p)$

Propositional ubiquitism: every truth is true everywhere: $\forall p(p \rightarrow \text{Everywhere } p)$

Propositional localism: some truth is not true everywhere: $\exists p(p \wedge \neg \text{Everywhere } p)$

4. Is propositional temporalism trivially true?

(1) Trump is president and Trump is not always president.

(2) $\exists p(p \wedge \neg \text{Always}(p))$ (1, EG).

A longer version of the above using ‘proposition’ talk.

(1) Trump is president and Trump is not always president.

(2) The proposition that Trump is president is such that: for it to be true is for Trump to be president.

(3) The proposition that Trump is president is such that always: it is true iff Trump is president.

(4) The proposition that Trump is president is such that: it is true and it is not always true.

(5) Some proposition is true and not always true.

But compare:

(1) People despise him and not every politician is such that people despise him.

(2) $\exists p(p \wedge \neg \text{every politician is such that } p)$

or

(1) People despise him and not every politician is such that people despise him.

(2) The proposition that people despise him is such that: for it to be true is for people to despise him.

(3) The proposition that people despise him is such that every politician is such that: it is true iff people despise him.

(4) The proposition that people despise him is such that: it is true and not every politician is such that it is true.

(5) Some proposition is such that: it is true and not every politician is such that it is true.

Lesson: if you’re a propositional eternalist, you should think that operators like ‘always’ work at least in this respect like quantifiers.

5. Easy arguments for propositional temporalism using propositional anaphora?

The apparent goodness of exchanges like the following pose a challenge for propositional eternalism:

(1) It’s raining — That’s true, but won’t be true tomorrow.

But we should be wary of such arguments since they threaten to prove too much:

- (2) It's raining — That's true, but isn't true where Sally is.
- (3) Movies like that make me fall asleep — That's true, but it isn't true for everyone.

So, what's going on? Suggestion: something similar to what's going on in the following examples:

- (4) I drove a Honda Accord for many years. — I used to drive that, but now I drive a Camry.
- (5) I drove the car the insurance company lent me for a month. — I only drove that for a week, then I bought a new car.
- (6) I drove the car the insurance company lent me for a month. — Alfred kept driving that as long as he could, but of course in his case it was a Lamborghini.
- (7) Fred had to replace the piece of plastic connecting the screen of his laptop to the keyboard. — I had to replace that too.

6. The temporalism/eternalism debate and the “specialness” of the present

What does this have to do with the question about the ‘privilege’ of the present instant?

Say that an instant of time is *accurate* iff all and only true propositions are true at it.

Here's an argument, using the terminology of ‘natural properties’ from Lewis's ‘New Work for a Theory of Universals’:

- (1) If propositional temporalism is true, the present time is the only accurate instant.
- (2) If propositional temporalism is true, accuracy is a highly *natural* property.
- (3) If propositional temporalism is true, most times are such that there isn't any even middlingly natural property that they uniquely instantiate.
- (4) So, if propositional temporalism is true, the present time is much more special than most times.

7. Does spacetime physics support propositional eternalism?

There's a certain picture of the world that many have taken to be supported by the success of spacetime field theories like GR in physics. Here's a first-pass (unduly specific in some ways, unduly noncommittal in others) attempt to say what it is:

The Manifold Picture: Concrete reality consists, fundamentally speaking, of objects called “spacetime points”. Their structure can be fully captured by an assignment of a four-dimensional differentiable structure (a criterion for what counts as a “smooth” co-ordinate system), together with a short list of ‘physically distinguished’ scalar, vector and/or tensor fields (e.g. the metric field, the electromagnetic field, the mass-density field) on this differentiable structure.

The picture has been widely thought, by both proponents and detractors, to support propositional eternalism (or the view that there is no deep contrast between time and space, which given propositional ubiquitousness entails propositional eternalism).

For today I'm less interested in discussing whether the picture is true, and more interested in discussing whether there is a good argument from it to Propositional Eternalism. Here's an attempt to make such an argument, where the first premise is supposed to (at least partially) articulate the picture:

Manifold Supervenience: Every true proposition is metaphysically necessitated by the collection comprising:

- (i) all truths about which spacetime points [and regions] there are
- (ii) all truths to the effect that a certain point or region is part of a certain other region
- (iii) all truths about the topological and differential structure of spacetime (i.e. which co-ordinate systems over the set of spacetime points are smooth)
- (iv) the truth, for each physically distinguished tensor field and spacetime point, about which tensor at that point is the value of that physically distinguished field
- (v) an appropriate "that's all" truth.

Perpetuity: Every necessary truth is always true.

Unchanging Spacetime: every truth of types (i)–(v) above is always true.

These together imply Propositional Eternalism. (If p is true, by Manifold Supervenience it's necessitated by some type (i–v) truths, so by Unchanging Spacetime it's necessitated by some eternal truths, so by Perpetuity it is itself eternal.)

There's also a different line of argument from the Manifold Picture to propositional eternalism, along the following lines:

- (1) If the Manifold Picture is true, instants of time are connected, spacelike, three-dimensional regions of spacetime.
- (2) If the Manifold Picture is true, then no connected, spacelike, three-dimensional region of spacetime is "special".
- (3) So, if the Manifold Picture is true, no instant of time is "special".
- (4) If propositional temporalism is true, the present is "special" (see previous section)
- (5) So if the Manifold Picture is true, propositional eternalism is true.

8. Do facts about propositional attitudes support propositional temporalism?

Consider this argument:

- (1) Every time at which the telescope is available is a time at which Arthur believes that Venus is visible.
- (2) The telescope is available.

(3) So, Arthur believes that Venus is visible.

It doesn't seem valid! The premises seem true, and the conclusion false, in the following scenario: Astronomer Arthur, knowing Venus is due to rise at 4am and to be visible until sunrise at 7am, arranged for the telescope to be available from 5am until 6am, in order to allow his students to make some observations of Venus. It is now 5.30am. But Arthur's watch has broken: he thinks it is just 2am. t

I hear a contrast here between (3) and (3'), which seems at least to have *an* interpretation on which it's true in the circumstances:

(3') Arthur believes that Venus is visible now.

This is all expected given propositional temporalism, hard to explain given propositional eternalism.

A common response: the semantic value of 'believe' isn't a relation between people and *propositions* but a relation between people and *properties of times*.

Compare verbs like 'want' and 'expect' that take an infinitival complement. Scenario: politician Jimmy has amnesia; he's watching TV, not realizing he is watching himself, and says 'That guy is going to win the election, whereas I'll be lucky if I ever get out of this hospital.' (4) seems unequivocally false, whereas (4') seems to have a true reading (perhaps along with some false ones):

(4) Jimmy expects to win the election.

(4') Jimmy thinks that he will win the election.

However saying that verbs like 'believe' don't take propositional arguments makes it hard to understand sentences like (5):

(5) It is both true [false / necessary / contingent...] and believed by Arthur that Venus is visible.

and challenging to understand the apparent validity of arguments like (6)–(8):

(6) Arthur believes Venus is visible.

(7) Venus isn't visible.

(8) So, Arthur believes something false [something that isn't the case].